smaller, particularly the upper teeth. Their roots reached deeper into the skulls in our ancestors, and they reduced over time with the harnessing of fire and the advancement of cooking. As humans began to consume an increasing amount of soft food, the teeth became smaller, but the lower jaw or the mandible did not shrink as rapidly, although it did, resulting in the protruding chin. Another reasoning is that with the easy availability of softer food, a reduction in teeth size was just more economical in terms of energy use. The problem with this hypothesis is something that you must have spotted already, there is no reason for the jawbone or the mandible to not shrink in size, in fact, it would offer a superior energy economy as well.

Along similar lines, another hypothesis is that the human face just became cuter over time in the interests of social cohesion. This can be comparable to how a domesticated dog looks in comparison to the wild wolf. In particular, a noticeable difference is a reduced snout. According to the hypothesis, humans domesticated themselves, and in the process, their evolution punched in their faces, leaving behind the chin. This theory has similar issues as the one before, as there is no good reason for the chin to reduce in lockstep with the rest of the face. As the midfaces of females are smaller than that of males, they should be expected to have proportionally larger chins, but that is not the case.

In primates, the face protrudes from the skull, whereas in humans it is kind of squished below the cranium, or the space that is used to house the skull. Primates also have a feature on their jawbones that humans do not. The front part of the mandible is connected by hard bone, and is known as a simian shelf. This simian shelf offers structural strength to the jawbone. The hypothesis is that as the human face got squished, and the Sapiens adapted to a bipedal gait, the airways would have gotten squished, if it were not for the chin that provides more space for the jaw. As the face got squished, the simian shelf simply moved from inside the jaw to the outside. The problem with this hypothesis is that the chin does not provide more space for the jaw, and the airways are squished anyway. In fact, the squished airways are suspected to have a lot to do with how we speak, and may in fact be an adaptation that makes it easier for humans to talk. A related theory is that chins are a result of sexual selection in animals, similar to the flanges on an orangutan. But the thing is only male orangutans have the flanges not the females. Typically, in most species that have an exaggerated body part for sexual selection, it is only one sex that displays the adaptation. This can range from the fans of male peacocks to the antlers of male stags. If sexual selection is the reason why humans have chins, it does not make sense for both sexes to get them.

Another theory put forth to explain chins is that they support the movement of the tongue, which is necessary for the complex speech of humans. As humans are the only species to talk, this hypothesis is a little difficult to test. Proponents of this hypothesis have supported their claims with simulations of the muscles required for speech with and without the chin. As far as the speech hypothesis is concerned, there is no evidence that the human jaw is expected to exert more force than other primates during their vocalisations, or during demanding periods such as suckling. Additionally, humans have a softer diet. Also, there is plenty of evidence to suggest that the other species of Homo were as conversant in their communication as Sapiens, and these species lacked a chin.

A male orangutan with the flanges at the cheeks

It might seem like a simple enough question, but it is one at the heart of an unlikely scientific controversy, heated debate, and active research. Scientists are using everything from genetics to computer simulations to come up with a definitive answer, but it is one that is always eluding them. Researchers have even put rabbits on soft diets to check if they evolve chins. Every time a scientist proposes a new theory, there are others who manage to demonstrate that it cannot possibly be the right explanation. In the end, it is a question that remains unanswered. Despite all the progress made by the human race and all the technologies at our disposal, nobody knows for sure why humans are the only animals on the planet with a chin.